Ben-Gurion University of the Negev

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Fourth Year Engineering Project

Preliminary Design

RF-Based Scanning and Localization System for SAR

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RF-Based Scanning and Localization System for SAR

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1.1 Abstract (English):

Rapidly locating missing persons in emergency scenarios (structure collapses, natural disasters, etc.) relies on the ability to scan large areas and detect human presence without line-of-sight. Many personal devices (e.g. Phones, Smartwatches, Earbuds) emit passive Wi-Fi and Bluetooth Low Energy (BLE) traffic (Probe, Beacon, Advertising), providing localization cues even in challenging environments. However, privacy protection mechanisms that randomize MAC addresses, alongside signal attenuation (path-loss, shadowing), and environmental noise hinder reliable mapping and consistent association of scans with the same transmitter over time.

The goal of this project is to develop a detection and localization system based on an airborne scanning unit, capable of handling randomized identities and providing precise positioning. The proposed method relies on a "Store & Forward" architecture. A passive scanning unit mounted on a drone (serving solely as a mobility platform), captures Wi-Fi frames (**with an optional extension to capture BLE advertising frames**) and extracts feature vectors (MAC, Received signal strength indicator (RSSI), occurrence rate, frame length, etc.) and tags them with spatial (GNSS) and temporal signatures. The data is stored locally on the unit during flight, while transmitting only a "Heartbeat" signal to the Ground-Station (laptop). Upon landing, the data is offloaded to the ground station, where a machine learning algorithm (Hybrid heuristic + logistic regression) for Re-identification (Re-ID) and statistical model for location calculation are applied. The core implementation and experimental evaluation in this work focus on Wi-Fi.

The expected outcome is the rapid generation of a probabilistic heatmap upon mission completion, indicating the location of missing people with high precision and facilitating the effective guidance of rescue forces.

Keywords:
BLE, Wi-Fi, GPS, RSSI, Re-Identification (Re-ID), MAC, Search and Rescue (SAR), Localization, Probabilistic Inference, Heat map.

1.2 Abstract (Hebrew):

איתור מהיר של נעדרים בתרחישי חירום (קריסת מבנים, אסונות טבע וכו') נשען במידה רבה על יכולת לסרוק תאי שטח נרחבים ולזהות נוכחות אנושית ללא הסתמכות על קו ראייה. מכשירים אישיים רבים (כגון טלפונים, שעונים חכמים, אוזניות) פולטים באופן פסיבי אותות Wi-Fi ובלוטוס עם צריכת אנרגיה נמוכה (BLE) (Probe, Beacon, Advertising) אשר יכולים לשמש לאיכון גם בתנאי שטח מאתגרים. עם זאת, מנגנוני הגנת פרטיות המבצעים רנדומיזציה לכתובות MAC, לצד דעיכת אות (path-loss, shadowing) ורעש סביבתי מקשים על מיפוי אמין ועל שיוך עקבי של סריקות לאותו משדר לאורך זמן.

מטרת הפרויקט היא לפתח מערכת איתור ולוקליזציה המבוססת על יחידת סריקה מוטסת, המסוגלת להתמודד עם זהויות אקראיות ולספק מיקום מדויק. השיטה המוצעת מתבססת על ארכיטקטורת "אגירה ושליחה" (Store & Forward). יחידת סריקה פסיבית שתותקן על גבי רחפן (כפלטפורמת ניידות בלבד) קולטת מסגרות Wi-Fi (עם אופציית הרחבה למסגרות BLE), מחלצת וקטורים של תכונות (MAC, RSSI, קצב הופעה, אורך מסגרת ועוד) ומתייגת בחתימת מיקום (GNSS) וזמן. המידע נשמר באופן מקומי על גבי היחידה במהלך הטיסה, תוך שידור אות חיים (Heartbeat) בלבד לעמדת הבקרה. לאחר הנחיתה, הנתונים נפרקים לעמדה הקרקעית (מחשב נייד) שם מופעלים אלגוריתם למידת מכונה לזיהוי מחדש (Re-ID)ומודל סטטיסטי לחישוב מיקום.

התוצאה הצפויה היא הפקה מהירה של מפת חום הסתברותית עם סיום המשימה. המצביעה בדיוק גבוה על אזורי הימצאות נעדרים ומאפשרת הכוונת כוחות חילוץ על פני שטח גדול.

מילות מפתח:

BLE, Wi-Fi, GPS, RSSI, Re-ID, MAC חילוץ והצלה, לוקליזציה, הסקה הסתברותית, מפת חום.

2. Project Goal:

The primary goal of this project is to minimize the Time-to-Locate (TTL) for Search and Rescue (SAR) teams operating in complex, non-line-of-sight (NLOS) environments. The project aims to bridge the operational gap between the presence of passive RF signals emitted by personal devices and the critical need for rapid, actionable spatial intelligence during emergencies.

To achieve this, the project seeks to resolve the identity and location uncertainty inherent in modern wireless traffic. By overcoming technical barriers such as MAC randomization, signal attenuation, and environmental noise, the system is designed to convert management frames (PCAP) + feature vectors (CSV) into a coherent common operating picture (COP). The proposed solution utilizes "Store & Forward" architecture to prioritize data integrity and processing depth, delivering a high-confidence probabilistic heatmap immediately upon mission completion.

The success of the project will be evaluated based on its ability to optimize signal processing performance, with the specific requirement of meeting the following quantitative acceptance criteria:

1. **Re-identification (Re-ID) consistency:** Maximize the integrity of frame clustering to effectively counter privacy mechanisms.
   Success criteria – achieve a Re-ID accuracy of at least 90% in correctly associating fragmented transmission sequences to a single logical entity, thereby preventing "ghost targets".
2. **Spatial Precision (Sharpness)**: Minimize the uncertainty area surrounding a target's estimated location to enable focused and efficient physical searches.
   Success criteria - The calculated uncertainty radius for a target shall be ≤ 10 meters, providing high-resolution localization relative to the search area.
3. **Statistical calibration reliability:** Ensure high statistical confidence in the generated probability map to prevent false negatives (miss detections).
   Success criteria – In at least 90% of test cases, the Ground Truth location of the device must fall within the boundaries of the calculated uncertainty circle presented to the operator.

3. Spec Sheet:

3.1 Project Name:

RF-Based Scanning and Localization System for SAR

3.2 Project Essence:

Our project leverages the fact that personal devices emit passive Wi-Fi and BLE signals without any human intervention. We utilize this property to gather location evidence by capturing those signals in motion and converting them into probabilistic heat map through rapid post-mission analysis.

Concretely, we build a passive scanning unit mounted on a drone acting as a mobility platform. Our solution employs Store & Forward architecture. Upon landing and data offloading, the ground station performs two main offline tasks: (1) Machine learning based device Re-Identification (Re-ID) to cope with MAC randomization. and (2) Statistical localization to estimate device positions. The minimal viable product and acceptance tests in this project are defined for Wi-Fi. BLE support will be implemented and evaluated only if the integration will be completed within the available time and resources.

3.3 Technology:

- **Sensing and Edge Compute:** Raspberry Pi Zero 2W acting as the air scanning unit. The system employs dual-interface architecture: (1) an external Wi-Fi dongle (wlan1) configured in Monitor Mode with channel hopping capabilities for passive sniffing, and (2) dedicated BLE dongle for beacon capture. A GNSS dongle provides synchronized positioning data.
- **Transport and Formats:** Store & Forward Architecture – To ensure data integrity and avoid packet lost due to link instability, heavy management frames (PCAP) and lightweight feature vectors are logged locally to the SD card in PCAP and CSV formats. Another Wi-Fi link (wlan0) is utilized exclusively as a Command & Control channel (C2), transmitting only periodic system health status (Heartbeat) to the operator and allowing operator to Start/ Stop the scan and download the files to the ground station.
- **Ground Station:** Offline Post-Mission Analysis Station – A python-based pipeline designed to ingest offloaded flight logs. The station manages a local database for storage and utilizes a Web UI for the visualization of the probabilistic heatmap and device tracking results upon mission completion.
- **Algorithms:** (1) *Re-ID* – Machine learning based association under MAC randomization, utilizing frame signature clustering (inspired by "Bleach"[1] and "Cappuccino[2]" methodologies) to resolve identity conflicts in the offline data. (2) *Localization* – Probabilistic grid likelihood estimation based on aggregated RSSI/ Path-loss models, refined via Weighted Least Squares (WLS) and Bayesian inference.

3.4 Research Stages:

1. Scanning Unit: Develop an End-to-End scanning unit (Pi Zero 2W, Wi-Fi + GNSS) that automatically executes START/STOP commands via a control link. The unit logs frames and feature vectors locally to onboard storage (SD card) and creates PCAP and CSV files for the ground station.

2. Re-ID Module: this module applies Hybrid heuristic + logistic regression algorithms to cluster fragmented transmissions from randomized MAC addresses into consistent, unique device identities.

3. Localization and Heat Map: implement a probabilistic inference engine to estimate device locations based on the reconstructed identities. The system will refine estimates using Weighted Least Squares (WLS) and Bayesian updates to rapidly generate a static heat map of high-probability zones immediately after data retrieval.

4. Field Experiments and Calibration: Conduct outdoor controlled tests with devices at known coordinates to validate the "Store & Forward" workflow, measure detection rates, quantify positioning error, and calibrate stochastic parameters (Path-loss exponent, Re-ID thresholds) against ground truth data.

3.5 Applications:

This system can serve SAR teams by converting passive Wi-Fi and BLE signals into a high-confidence probabilistic heat map that directs search efforts.

Deployed in the field, it facilitates evidence-based prioritization of search sectors, significantly shortening the Time-to-Locate (TTL) and supports clearer operational decisions. The modular design can support parallel scanning with multiple units, enabling wider coverage and faster Common Operating Picture (COP).

3.6 System Performance Specification:

The system is designed to passively detect Wi-Fi and BLE emissions from personal devices in the area, log the data locally, and support rapid post-mission analysis. The ground station performs Re-ID under MAC randomization, estimates a per-ID probabilistic localization (uncertainty circle) from RSSI samples and generates a heatmap to guide SAR teams over a 100x100 m search area.

Core Performance targets:

- **Re-ID quality** - Achieve Re-ID accuracy of at least 90% in clustering randomized MAC addresses unique logical identities.
- **Localization quality** -
    - Calibration – in at least 90 % of cases, the device's real location shall fall within the calculated uncertainty circle
    - Sharpness – The calculated uncertainty radius shall be <= 10 meters
    - Point accuracy –The distance between the estimated location and the ground truth shall be < 10 meters

Components:

Component	Quantity	Price [$]
Raspberry Pi Zero 2W	1	26
Wi-Fi Dongle	1	10
BLE Dongle	1	16
GNSS Dongle	1	12
USB Hub	1	10
Total per Unit		74

3.7 System Scheme:

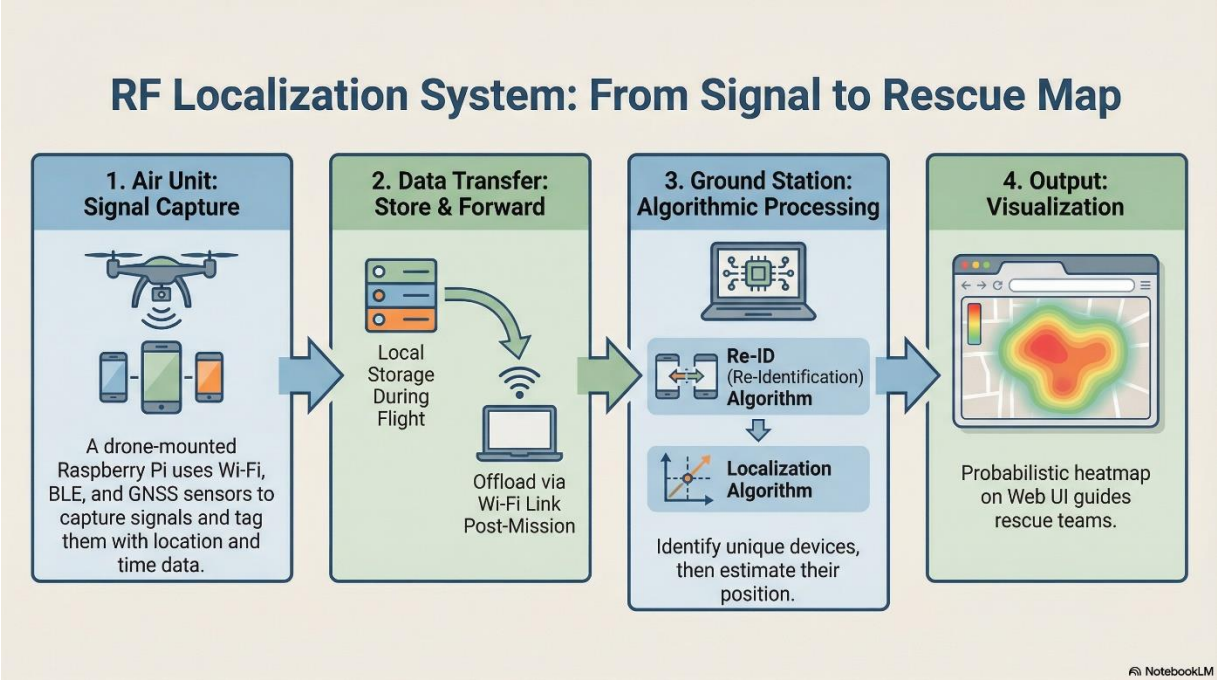


Figure 1 – System Scheme

4. Brief Literature Review:

The goal of developing a system to detect and localize missing persons via wireless signals relies on overcoming major challenges posed by the complexity of radio frequency (RF) environments and modern privacy protocols. The two primary technical challenges addressed by recent academic and commercial efforts are Re-Identification (Re-ID) of transmitters and Localization of passive RF emitters.

4.1 Re-Identification (Re-ID): Overcoming MAC Randomization

Many personal devices emit passive Wi-Fi and Bluetooth Low Energy (BLE) traffic (such as Probe, Beacon, and Advertising frames). However, to protect user privacy and location tracking, modern devices implement MAC address randomization, causing the MAC address to change randomly and unpredictably over time. This mechanism disrupts the continuity and semantics of probe requests, leading to fragmented data from the same device.

Re-ID methods aim to associate these randomized MAC addresses back to a single device identifier. These methods rely on extracting device-specific "signatures" from frame attributes that remain stable despite the randomized address, including Sequence numbers (SEQ), Information elements (IEs) and Received Signal Strength Indicator (RSSI) values.

Bleach[1]: Logistic Regression for Conflict Resolution

Bleach[1] is a framework that models the MAC association problem as the resolution of conflicts that arise in small time intervals when a randomized MAC address disappears and a new one appears. Bleach[1] employs a novel association algorithm utilizing **logistic regression.**

The framework uses two primary signature types to resolve conflicts:

1. **Time-based signature:** This is the mean inter-frame time, calculated across probe request bursts. The underlying principle is that the frequency of sending probes during an active scan is unique to a device and highly consistent.
2. **Frame-based signature:** This relies on the IEs contained in the probe request frame, which depict the device's abilities. Bleach[1] identifies the top eight most frequent IE elements (such as SSID, Supported Rates, HT capabilities) as the basis for the signature.

For MAC association, Bleach[1] trains the logistic regression model on two features derived from these signatures: IE similarity (f_1) and the absolute difference in mean inter-frame time (f_2). Finally, they use the Sequence Number Gap (d_{seq}) as a final filter; the method establishes a threshold (SEQ_{TH}) of 64, based on the observation that approximately 85% of correctly associated trails exhibit a gap smaller than 64. Evaluations shows that Bleach[1] achieves robust association accuracy with results as high as 99% in controlled scenarios.

Cappuccino[2]: Multi-modal Deep Learning and Network Flow

Cappuccino[2] proposes a sophisticated approach that leverages self-supervised learning and multiple modalities to accurately associate frames in complex, crowded scenarios. This

framework consists of two main modules: a **frame correlation estimator** and a **multi-frame association** module.

1. **Frame Correlation Estimator:** This component estimates the **pairwise correlation probability ($\boldsymbol{P(\varphi_{ij})}$)** between any two frames P_i and P_j. It uses a neural network with a **Siamese structure** trained via **contrastive learning** to jointly consider multiple modalities without requiring manual data labeling. The modalities include:

   IEs: Encoded using **Sequential Encoding** to capture long-range dependencies with the IE bit sequence.
   SEQ: Processed using **modulus subtraction** to capture interval patterns

   RSSI: Encoded into a compact latent space using **Spatial Encoding** (fully connected layers) to reflect user movement.

2. **Multi-frame Association:** To alleviate the ambiguity of "association conflict" in crowded environments, Cappuccino[2] formulates the multi-frame association problem as a **minimum-cost flow optimization** problem over a batch of probe requests. The cost of an edge in the network flow graph is defined as the negative log of the frame correlation probability ($-log\boldsymbol{P(\varphi_{ij})}$). Minimizing this cost yields the globally optimal association for all frames in the batch.

Cappuccino[2] achieves high precision in clustering, showing V-measure scores > 0.85 and discrimination > 80% accuracy.

<u>4.2 Localization:</u>

Once MAC addresses are consistently associated via Re-ID, the localization challenge involves converting the detected signal strength (RSSI) into a reliable positional estimate.

Localization Core: The process begins by converting RSSI samples to range likelihoods using **log-distance path-loss model**. These range likelihoods are fused across GPS-tagged vantage points via Weighted Least Squares (WLS) and smoothed with a Bayesian filter. The fused surface is rendered as a probabilistic heat-map, and the Maximum-a-Posteriori (MAP) (center of each radius) is the estimated device location. The system aims for a localization quality where 90% of real locations are within their calculated uncertainty circles with a sharpness of < 10m.

Krypto[3]: Search Path Optimization

The Krypto[3] system demonstrated the use of a UAV (Phantom 2) to assist SAR operations by

detecting wireless signals, primarily Wi-Fi, from mobile phones. Krypto's [3] localization approach used **Trilateration** and **Dual RSSI Trend Based Localization (DuRT).** The system assumes the UAV is the mobility platform and emphasizes the role of the searching path in minimizing location errors.

Studies showed that the localization accuracy is dependent on the relative position of the sensing points; location error is minimized if the sensor points are located in at least three quadrant planes around the target signal source. Krypto[3] analyzed several search paths, finding that complex paths like the **4-Leaf** structure provided better coverage and lower location error compared to simpler patterns like the Single Circle, while consuming less power than the Space-filling Curve.

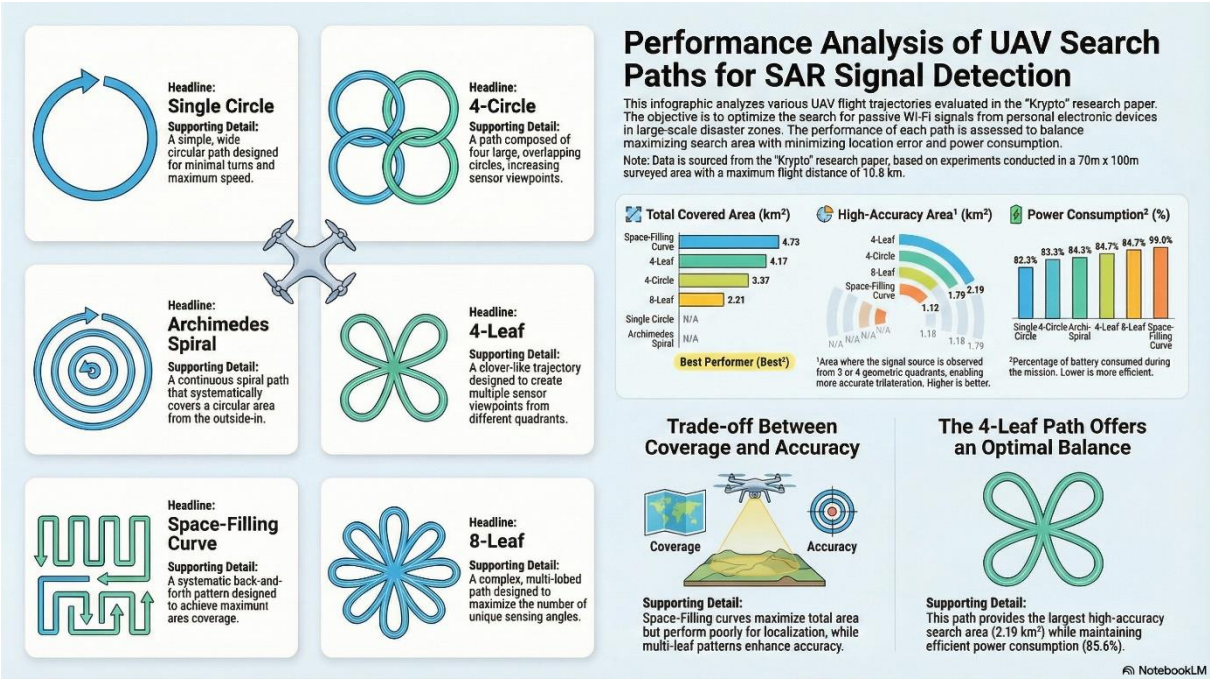


Figure 2 – Analysis of search path based on Krypto [3]

BlueFly[4]: Real-Time Detection and Visualization

BlueFly[4] is a state-of-the-art commercial detection device designed to enhance SAR missions by detecting unintended, low-power BLE and Wi-Fi signals from personal and medical devices (like pacemakers and glucose monitors). BlueFly[4] validates the passive RF localization approach in practice by operating independent of the drone's power or communication system, making it adaptable to several drone types. A key feature is its real-time data visualization: collected data is transmitted immediately to Team Awareness Kit capable Android devices, providing teams on the ground with updates on detected signal locations.

5. Proposal Solution Plan:

This section details the system architecture, hardware specifications, and algorithmic strategies designed to minimize Time-to-Locate (TTL) in Non-Line-of-Sight (NLOS) environments.

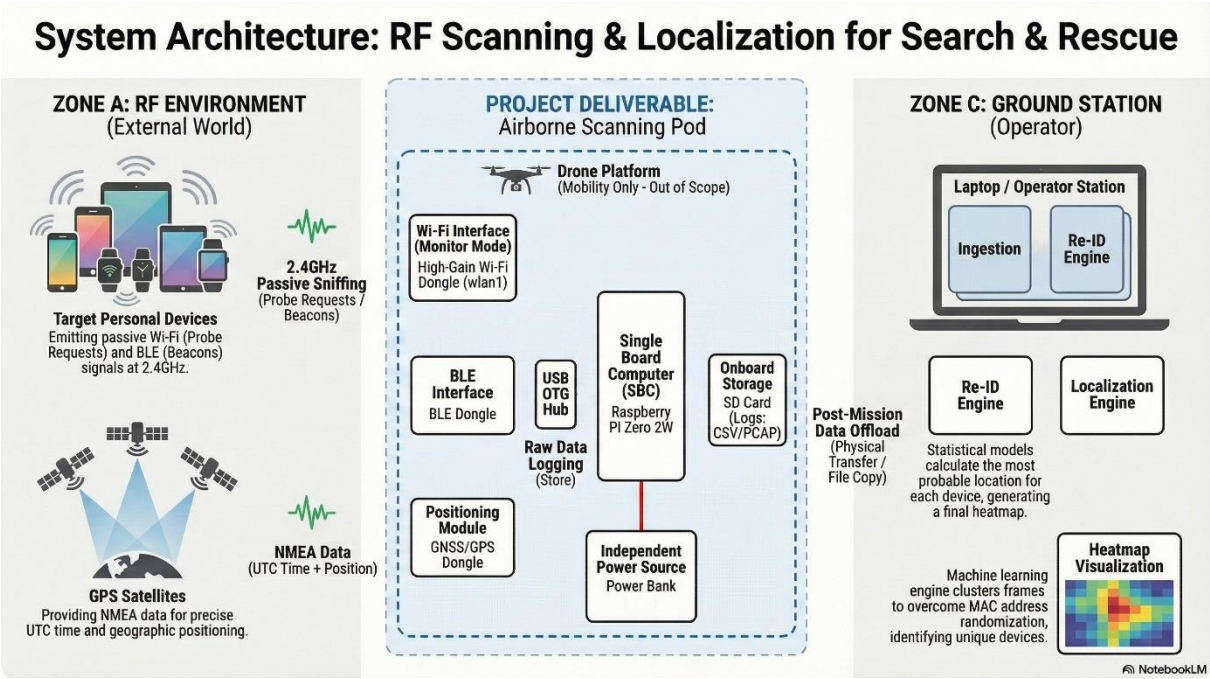


Figure 3 – System Architecture

System Architecture:

The system operates in a three-stage pipeline, as illustrated in Figure 3:

1. **Acquisition:** The drone-mounted Air Unit passively captures Wi-Fi and BLE management frames, tagging each observation with a synchronized UTC timestamp and GNSS coordinate.

2. **Storage:** Data is logged locally to an onboard SD card during flight. A lightweight "Heartbeat" signal is transmitted via a C2 link for system health monitoring.

3. **Analysis:** Upon landing, logs are offloaded to the Ground Station for offline processing. The station executes a Re-Identification (Re-ID) engine to resolve randomized MAC addresses and a Probabilistic Localization engine to generate a high confidence heatmap to guide SAR teams.

Hardware architecture: The airborne payload is designed as a modular, platform-independent "pod" powered by an independent power source.

- **Processing Core:** Raspberry Pi Zero 2W [10], selected for its balance of low power consumption and sufficient I/O throughput for writing high-velocity PCAP streams.

- **RF Interface (Data Acquisition):**
    - Wi-Fi interface: External Wi-Fi dongle configured in **Monitor Mode** (wlan1) with channel hopping across 1, 6, 11 to sweep the 2.4GHz band, equipped with High-Gain antenna for capturing management frames (Probes/Beacons).
    - BLE interface: dedicated USB sniffer for capturing BLE advertising / scanning frames.
- **Command & Control (C2) Interface:** Onboard Wi-Fi interface (wlan0) configured as an Access Point (AP) or Client for Command and Control.
- **Positioning:** GNSS USB Dongle [11] providing NMEA stream data for spatial tagging and UTC time synchronization.
- **USB Hub:** A compact USB hub to connect the Wi-Fi dongle, BLE dongle, and GNSS dongle to the Pi Zero 2W.
- **Local Storage:** SD card storage for generating CSV + PCAP logs locally during flight.
- **Power Source:** Independent power bank supplying the Air Unit to maintain platform independence from the UAV.

Software Architecture: The software stack is divided into two distinct environments:

Airborne Service:

- **Capture Pipeline:** Utilizes "tcpdump" subprocesses to capture IEEE 802.11 management frames.
- **Data Logging:** Implements a dual-logging mechanism:
    - **PCAP:** Raw 802.11 management frames for deep forensic analysis.
    - **CSV:** Lightweight feature vectors (MAC, RSSI, GPS Lat/Lon/Alt, Sequence Control) for rapid algorithmic processing and sync GNSS with PCAP.
- **C2 Server (Control Plane):** A **FastAPI** application (control_app.py) managing a **WebSocket** endpoint. This allows the ground operator to remotely trigger scan start/stop commands and monitor system health (CPU, GPS Fix status) in real-time without transmitting the heavy data payload.

Ground Station Analysis Suite:

- **Ingestion Module:** Fetches logs via HTTP/Physical transfer.
- **Algorithmic Core:** Implements Re-ID (clustering) logic and Probabilistic Localization engines.
- **Visualization:** Generates a geo-referenced heatmap overlaying the probability of victim presence on a satellite map.

Store & Forward Concept:

- During flight, all captured observations are written to local storage on the Air Unit to avoid reliance on continuous connectivity.
- After landing, logs are offloaded to the Ground Station over a Wi-Fi link (or physically, if required).
- This design prioritizes data completeness and robustness over real-time streaming.

Algorithmic Logic: The core intelligence of the system lies in the post-mission processing, which converts raw, noisy RF data into actionable spatial intelligence.

1. Re-Identification (Re-ID) Engine – Objective: To resolve the identity of devices employing MAC Address Randomization. Modern devices periodically rotate their MAC addresses, fragmenting the data trail. The Re-ID engine clusters these fragments into unique logical entities.

Logic: The system utilizes a **Hybrid Approach** combining heuristic filtering with Machine Learning, inspired by the **Bleach[1]** and **Cappuccino[2]** frameworks.

Stage A: Heuristic Filtering (Sequence Number Analysis): Rationale - While MAC addresses change, the 12-bit Sequence Control field in IEEE 802.11 frames often behaves as a linear counter (modulo 4096) that persists across MAC rotations for short bursts [9].

Mathematical Model: The algorithm calculates the gap (d_{seq}) between the last frame of a disappearing trail (T_{old}) and the first frame of a new appearing trail (T_{new}).

$$d_{seq} = (SEQ_{new} - SEQ_{old}) mod 4096$$

- **Where:**
  - SEQ_{new}: The sequence number of the first packet in the new candidate trail.
  - SEQ_{old}: The sequence number of the last packet in the previous trail.
  - $mod4096$: The modulo operation handles the counter rollover.

Decision Logic:

$$Match = \begin{cases} True, & if\ 0 < d_{seq} < \delta_{th}\ AND\ \Delta t < t_{max} \\ False, & otherwise \end{cases}$$

- **Variable Breakdown:**
  - δ_{th}**:** Sequence gap threshold (e.g., δ_{th}≈ 64 as in Bleach [1]).
  - Δt**:** Time difference between the last frame in T_{old} and the first frame in T_{new}.
  - t_{max}**:** Maximum allowed time window for a valid rotation event.
- **Why this formula?** Empirical studies (e.g., *Bleach*[1]) indicate that legitimate MAC rotations occur within short timeframes. A small, positive gap ($\delta_{th} \approx 64$) strongly implies continuity of the same hardware counter, distinguishing it from a random collision with a different device.

Stage B: Machine Learning Classification (Conflict Resolution) To resolve collisions where multiple devices exhibit similar sequence gaps, the system employs a **Logistic Regression** classifier. This model evaluates a feature vector (f) based on time and frame content signatures [8]:

- **Time-based Signature (f_1):** The difference in Mean Inter-Frame Spacing (IFS) between bursts.
- **Frame-based Signature (f_2):** Similarity score of Information Elements (IEs) such as Supported Rates and HT Capabilities.
- The classifier outputs a probability score (P_{match}), associating the randomized trail only if $P_{match} > 0.9$.

2. Probabilistic Localization Engine - Objective: To estimate the device location based on Received Signal Strength Indicator (RSSI) samples collected from multiple vantage points.

Step A: Range Estimation (Log-Distance Path Loss) [5] The system converts RSSI samples into distance likelihoods using the standard Log-Distance Path Loss model:

$$RSSI(d) = P_{tx} - 10 \cdot \eta \cdot log_{10}(d) + X_{\sigma}$$

- **Variable Breakdown:**
  - $RSSI(d)$: The received signal power at distance d (measured in dBm).
  - P_{tx}: The reference transmission power at 1 meter (typically calibrated to -30 to -40 dBm for smartphones).
  - η - (Path Loss Exponent): An environmental constant representing signal decay rate. For SAR scenarios (outdoor/mixed), we set $\eta \approx 2.5 - 3.0$.
  - d: The Euclidean distance between the drone and the target (meters).
  - X_σ: A zero-mean Gaussian random variable representing shadowing noise ($\sigma \approx 4 - 6\, db$).
- **Why this formula?** This is the standard RF engineering model for relating signal attenuation to distance. Unlike geometric triangulation (which fails with noisy RSSI), this probabilistic model accounts for the stochastic nature of signal fading in complex environments (X_σ).

Step B: Sensor Fusion & Bayesian Grid Update [6] To mitigate the noise (X_σ), the system employs Weighted Least Squares (WLS) to find an initial geometric centroid [7], followed by a Bayesian Grid Update. Mathematical Model: The search area is divided into a grid of cells C_i. The probability $P(C_i)$ of a target being in cell C_i is updated for each new measurement M_k:

$$P(C_i|M_k) = \frac{P(M_k|C_i) \cdot P(C_i)}{\sum_j P(M_k|C_j) \cdot P(C_j)}$$

- **Where:**
  - $P(C_i)$: The prior probability (probability before the new measurement).
  - $P(M_k|C_i)$: The **Likelihood Function**, derived from the Path Loss model - In practice, $P(M_k \mid C_i)$ is modeled as a Gaussian likelihood [5] over the RSSI residual between the measured M_k and the expected RSSI at the distance from cell C_i, using the calibrated σ. (probability of seeing RSSI M_k if the target is truly at C_i).
- **Why this formula?** Bayesian inference allows the system to incrementally improve accuracy. As the drone collects more samples from different angles, "ghost" locations fade out, and the true location converges, effectively filtering out measurement noise.

5.1 Project Constraints:

- **Physical and Systemic Independence:** The scanning unit must remain lightweight and low-power to be compatible with a drone acting as a mobility platform.
- **Hardware and Edge-Compute Limitations:** The Air Scanning Unit, based on the Raspberry Pi Zero 2W has limited processing and memory resources, therefore the software must be lightweight and efficient, prioritizing data collection over data processing.
- **RF Environment Complexity:** Signal metrics such as RSSI are highly susceptible to multi-path fading and non-line-of-sight (NLOS) conditions, this necessitates multi-point data collection strategy and extensive field calibration to ensure raw data is converted correctly into actionable location information.
- **Privacy Protocols:** Modern device implementations of MAC address randomization limits the continuity of captured data, requiring specialized Re-ID algorithms to maintain device tracking.

5.2 Project Assumptions:

- **Sufficient Passive Emissions:** It is assumed that the target devices emit Wi-Fi or BLE traffic periodically, such as Probe Requests or Advertising frames, with a high enough frequency to allow statistical sampling throughout the drone's flight path.
- **Accurate Spatial and Temporal Tagging:** The GNSS dongle must provide synchronized GPS data with sufficient accuracy and timestamp consistency to allow accurate RSSI samples from different vantage points.
- **Localization Model Validity:** The localization approach is assumed to be calibratable to the operational environment, yielding statistically stable results.
- **Feature Extractability:** The air-unit, operating in monitor mode, must be able to reliably capture relevant frames and extract consistent signatures necessary for Re-ID such as IEs and SEQ from the PCAP files.

5.3 Initial Defined Risks:

- **Re-ID Failure:** High device density or rapid randomization may cause fragmented data, leading to incorrect clustering or "ghost targets".

Mitigation Strategy: Implementing multi-modal signatures (SEQ, IEs and timings) inspired by the “Bleach”[1] and “Cappuccino”[2] frameworks to ensure at least 90% association accuracy.

- **Unstable Localization Accuracy:** Non-line-of-sight (NLOS) conditions and environmental noise cause RSSI volatility, potentially exceeding the 10m error threshold.

  Mitigation Strategy: Utilizing multi-point WLS fusion and Bayesian filtering to smooth estimates, combined with rigorous field calibration of stochastic path-loss parameters.
- **Data Integrity and System Failure:** Limited I/O capacity or software crashes on the Raspberry Pi may result in corrupted or incomplete PCAP/CSV logs.

  Mitigation Strategy: Adopting a “Store & Forward” architecture to prioritize local disk writes over transmission and using ‘Heartbeat” signal to monitor system health via a dedicated Command & Control channel.
- **Sparse Signal Emission:** Devices may emit RF traffic too infrequently for effective statistical sampling during a fast-moving drone flight.

  Mitigation Strategy: Expanding the capture scope to include multiple frame types and designing flight paths that maximize airtime over high-probability search zones.
- **GNSS Inaccuracy:** Poor satellite fix or signal drift in complex environments may lead to incorrect spatial tagging of captured RF traffic.

  Mitigation Strategy: Monitoring the GPS fix via the Heartbeat signal and using high-precision GNSS module to ensure reliable coordinates.

5.4 Project's Content Define:

Detailed hardware/software architecture and mathematical models are described in Section 5 (Proposal Solution Plan). This section summarizes scope, deliverables, interfaces, and boundaries.

A. Project Deliverables

The project encompasses three primary development vectors: Hardware integration, Airborne Embedded Software, and Ground Station Analysis Software.

1. Hardware Deliverable: Independent Air Scanning Unit: A modular "Edge Unit" designed for mounting on a commercial UAV (mobility platform).

2. Software Deliverable: Airborne Intelligence Service: A Python-based system service (wifi_sniffer_logger.py and control_app.py) running on the airborne unit.

3. Software Deliverable: Ground Station Analysis Suite: An offline processing pipeline deployed on the operator's laptop.

B. Algorithmic Logic & Mathematical Models

The system employs two core algorithmic engines to convert raw RF signals into actionable location intelligence – Re-ID Module and Localization Module.

C. System Interfaces

The system interfaces are designed to separate critical data logging from operational control, ensuring robustness.

1. RF Input Interface (Passive)

- **Mechanism:** The wlan1 interface operates in Monitor Mode, utilizing channel hopping (Channels 1, 6, 11) to sweep the 2.4GHz spectrum.
- **Data Flow:** Unidirectional (Ingress only). No frames are transmitted on this interface to maintain silence and save power.

2. Command & Control (C2) Interface

- **Mechanism:** A persistent Wi-Fi link (wlan0) connects the Air Unit to the Ground Station.
- **Protocol:** High-level HTTP and **WebSocket** communication via the **FastAPI** framework.
- **Functionality:**
    - **Live Status:** The "ScanManager" class pushes JSON updates ("status": "running", "gps_fix": 1) via WebSocket to the operator's dashboard.
    - **Control:** The operator sends POST requests (/scan/start, /scan/stop) to trigger the logging service.

3. Data Offload Interface (Post-Mission)

- **Mechanism:** HTTP File Transfer (Wireless Offload).
- **Process:**
    1. Upon landing, the operator navigates to the "Files" tab in the Web UI.

2. The browser requests the file list via GET /logs.
3. The Ground Station downloads the high-density CSV/PCAP files via GET /logs/{filename}.

- **Rationale:** This eliminates the need for physical disassembly to retrieve the SD card, while maintaining the "Store & Forward" reliability (files are generated locally first, preventing data loss during flight).

D. Scope & Boundaries

In-Scope:

- Development of the Airborne Pod (Hardware + Embedded Python Logic).
- Implementation of the Ground Station Web UI and Analysis Algorithms.
- Integration of COTS (Commercial Off-The-Shelf) GPS and Wi-Fi dongles.
- Field testing for calibration of the Path Loss Exponent (η).

Out-of-Scope:

- **Drone Development:** The drone is treated strictly as a carrier payload. The project does not involve flight controller programming or aerodynamics.
- **Real-Time Streaming:** Heavy data is not streamed in real-time. The operational concept relies on rapid post-mission analysis.
- **Decryption:** The system analyzes only unencrypted Management Frames (Headers). No payload decryption or Deep Packet Inspection (DPI) of user data is performed.

5.5 Set of Final Tests Proposal:

The final testing phase is designed to validate the system's Key Performance Indicators (KPIs) through a rigorous series of 20 field trials (runs). To account for the lack of external reference equipment (e.g., DGPS) and the inability to access internal device logs due to privacy protocols, the tests will rely on pre-surveyed static locations measured by the system's own GNSS module. This ensures that measured errors reflect the algorithmic localization performance relative to the sensor's coordinate system.

Test Setup:

1. **System Under Test:** Air Scanning Unit based on Raspberry Pi Zero 2W, utilizing a Wi-Fi monitoring interface configuration. The system operates in a Store & Forward architecture, logging management packets and feature vectors locally to CSV/ PCAP formats for offline post-mission analysis.
2. **Test Environment:** An open outdoor area (approx. 100X100m) characterized by minimal RF interference to isolate system performance variables.
3. **Test Targets:** 1-5 standard smartphones (iOS/Android) with Wi-Fi interface enabled. Devices will remain with screens off.
4. **Ground Truth Acquisition:** Prior to flight operations, specific waypoints will be measured and marked using the Air Scanning Unit GNSS dongle (averaged over 2-minute stationary period). These coordinates serve as the relative Ground Truth for accuracy calculations.

Detailed Test Runs: (Total: 20 Runs) – The testing protocol is divided into three quantitative groups.

Group 1: System Integrity & Data Continuity (3 Runs): Verify the robustness of the "Store & Forward" architecture and ensure zero data loss during the logging process.

Run 1 (Static) – Ground operation for 10 minutes to validate thermal stability and log generation.

Run 2 (Flight) – A full flight with active channel hopping to verify that CPU load does not result in packet drops or log corruption.

Run 3 (Power Loss) – Induced power loss scenario to validate file system recovery and ensure previous data is not corrupted.

Acceptance Criteria:

File Integrity – 100% of generated CSV and PCAP logs must be readable and structurally valid.

GNSS Synchronization - >95% of logged packets must contain a valid GPS fix.

Group 2: Re-Identification Efficiency (7 Runs): Validate the algorithm's ability to cluster fragmented transmissions from randomized MAC addresses into unique logical identities.

Methodology – Due to inability to correlate specific randomized MACs with ground truth without internal logs, the test employs a counting heuristic. N distinct smartphones are placed

at widely separated static locations (>20m apart) for 15 minutes.

Runs 4-10 – The drone executes a flight over the area for 15 minutes, The offline processing pipeline calculates the number of unique targets detected.

Acceptance Criteria:

Target Accuracy – The system shall report N (± 1) unique targets in at least 6 out of 7 runs.

Group 3: Localization Accuracy & Calibration (10 Runs): Quantify the spatial positioning error of the probabilistic heatmap relative to the pre-surveyed markers.

Runs 11-15 (Line-of-Sight – LOS): Targets placed on elevated markers with clear RF propagation paths.

Runs 16-20 (Non-Line-of-Sight – NLOS): Targets concealed under foliage or physical obstacles to simulate realistic SAR scenarios.

Measurement – Calculation of the Euclidean distance between the estimated Maximum A-Posteriori (MAP) coordinate and pre surveys ground truth.

Acceptance Criteria:

Point Accuracy – The mean Euclidean error across all runs shall be <10 meters.

Statistical Calibration – In ≥ 90% of the runs, the true location of the device must fall within the calculated uncertainty radius.

5.6 Budget Estimation:

1. **Salaries:**

	Hours per week	Weeks	Cost per Hour [$]	Total cost [$]
Student Hours	10*2=20	30	25	15000
Supervisor Hours	1	30	100	3000
Consultant	1	4	120	480
Total +25%				23100

2. **Components:**

Component	Quantity	Price [$]
Raspberry Pi Zero 2W	1	26
Wi-Fi Dongle	1	10
BLE Dongle	1	16
GNSS Dongle	1	12
USB Hub	1	10
Total per Unit		74

3. **Special Equipment:**

	Cost	Time	Total cost [$]
UAV Usage	100 per hour	12 hours	1200
Cloud Computing	TBD	TBD	
LLM	20 per month	8 months	160
Total			1360

4. **Other:**

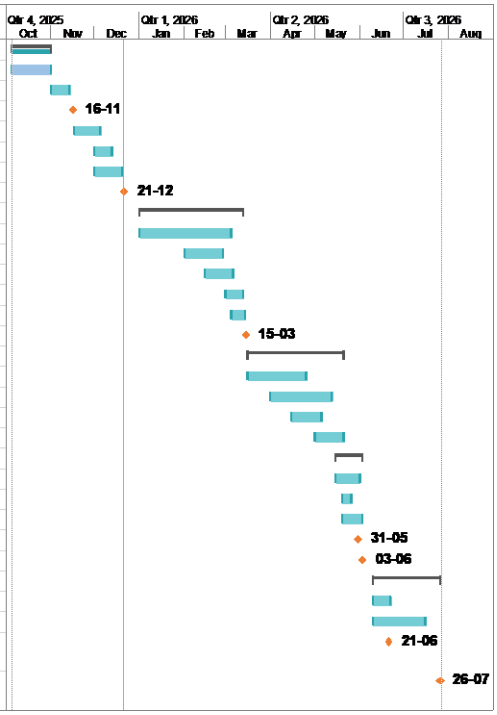
	Quantity	Price [NIS]
Cables	2	3*2=6
Soldering KIT	1	30
Soldering paste	1	5
Total		41

5. **Total:**

	Price [$]
Salaries	23100
Components	74
Special Equipment	1360
Other	40
Total	24574

5.7 Schedule and Devision of Work:

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors
1			Phase 1 Initiation & Planning	21 days	Sun 05-10-25	Fri 31-10-25	
2			Litrature Review & Concept	21 days	Sun 05-10-25	Fri 31-10-25	
3			Write PDR Document	10 days	Sat 01-11-25	Thu 13-11-25	
4			[MILESTONE] PDR Submission	0 days	Sun 16-11-25	Sun 16-11-25	
5			Procurement & Setup	14 days	Mon 17-11-2	Thu 04-12-25	
6			HW Hello World (Monitor Mode)	10 days	Mon 01-12-2	Fri 12-12-25	
7			Write Preliminary Report	15 days	Mon 01-12-2	Fri 19-12-25	
8			[MILESTONE] Preliminary Submissio	0 days	Sun 21-12-25	Sun 21-12-25	
9			Phase 2 Core Development	51 days	Thu 01-01-2(	Thu 12-03-2(	
10			Dev Airborne Embedded Service	45 days	Thu 01-01-26	Wed 04-03-2	
11			Dev Hardware Pod Integration	20 days	Sun 01-02-26	Thu 26-02-26	
12			Dev Ground Station (Basic)	15 days	Sun 15-02-26	Thu 05-03-26	
13			Integration Test 1 (Store & Forwa	10 days	Sun 01-03-26	Thu 12-03-26	
14			Prepare Progress Presentation	7 days	Thu 05-03-26	Fri 13-03-26	
15			[MILESTONE] Progress Submission	0 days	Sun 15-03-26	Sun 15-03-26	
16			Phase 3 Algorithm Implementation	48 days	Mon 16-03-2	Wed 20-05-2	
17			Dev Re-ID Engine (ML Logic)	30 days	Mon 16-03-2	Fri 24-04-26	
18			Dev Localization Engine (Bayesiar	30 days	Wed 01-04-2	Tue 12-05-26	
19			Field Test Group 2 (Re-ID)	15 days	Wed 15-04-2	Tue 05-05-26	
20			Field Test Group 3 (Localization)	14 days	Fri 01-05-26	Wed 20-05-2	
21			Phase 4 Conference Preparation	13 days	Fri 15-05-26	Wed 03-06-2	
22			Write Abstracts & Poster Design	12 days	Fri 15-05-26	Sun 31-05-26	
23			Data Analysis & Tuning	4 days	Wed 20-05-2	Mon 25-05-2	
24			Prepare Conference Presentation	10 days	Wed 20-05-2	Tue 02-06-26	
25			[MILESTONE] Abstracts & Poster	0 days	Sun 31-05-26	Sun 31-05-26	
26			[MILESTONE] Project Conference	0 days	Wed 03-06-2	Wed 03-06-2	
27			Phase 5 Project Closure	33 days	Wed 10-06-2	Sun 26-07-2(	
28			Write Final Presentation	9 days	Wed 10-06-2	Sun 21-06-26	
29			Write Final Book (Report)	26 days	Wed 10-06-2	Wed 15-07-2	
30			[MILESTONE] Final Presentation Upload	0 days	Sun 21-06-26	Sun 21-06-26	
31			[MILESTONE] Final Report Submission	0 days	Sun 26-07-26	Sun 26-07-26	



6. References:

[1] A. K. Mishra, A. C. Viana, and N. Achir, “Bleach: From Wifi Probe-Request Signatures to Mac Association”, Ad Hoc Networks, vol. 164, art. 103623, 2024.

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[3] Y.-H. Ho, Y.-R Chen, and L.-J. Chen, “Krypto: Assisting Search and Rescue Operations using Wi-Fi Signal with UAV”, in Proc. ACM DroNet, 2015, pp. 3-8.

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[7] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*. Upper Saddle River, NJ, USA: Prentice Hall, 1993, ch. 3.

[8] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009, ch. 4.

[9] IEEE Standard for Information Technology—Telecommunications and Information Exchange Between Systems—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications, IEEE Std 802.11-2020, 2021.

[10] Raspberry Pi Ltd., *Raspberry Pi Zero 2 W Product Brief*, 2021.

[11] u-blox AG, *MAX-M10 Standard Precision GNSS Modules Data Sheet*, UBX-21044856, 2021.

7. Attachments:

12.2 טופס כיבוד זכויות יוצרים וסודיות

יש לצרף את הטופס הבא חתום לדוח מכין:

אני מצהיר שלא אעשה שימוש בפרויקט ההנדסי שלי בכל חומר בעל זכויות יוצרים כגון:

טקסט,

תמונה,

אודיו,

וידיאו ,

מוזיקה,

סרט,

אנימציה,

תוכנה

חומרה

תיכנון מעגל

ללא קבלת אישור מראש מבעל הזכויות

אני מצהיר שאשלב בפרויקט ההנדסי שלי בדוחות, סרטונים, והרצאות אינפורמציה שאינה נחלת הכלל רק בתנאי שאושרה מראש ע"י בעל הזכויות.

הרישום לפרויקט ההנדסי משמש ההתחייבות שלי לקיים ולכבד זכויות יוצרים וסודיות

חתימה: רון אשכנזי _____ תאריך: 19/12/2025

12.2 טופס כיבוד זכויות יוצרים וסודיות

יש לצרף את הטופס הבא חתום לדוח מכין:

אני מצהיר שלא אעשה שימוש בפרויקט ההנדסי שלי בכל חומר בעל זכויות יוצרים כגון:

טקסט,

תמונה,

אודיו,

וידיאו ,

מוזיקה,

סרט,

אנימציה,

תוכנה

חומרה

תיכנון מעגל

ללא קבלת אישור מראש מבעל הזכויות

אני מצהיר שאשלב בפרויקט ההנדסי שלי בדוחות, סרטונים, והרצאות אינפורמציה שאינה נחלת הכלל רק בתנאי שאושרה מראש ע"י בעל הזכויות.

הרישום לפרויקט ההנדסי משמש ההתחייבות שלי לקיים ולכבד זכויות יוצרים וסודיות

חתימה:_____ תאריך:_____19/12/2025

המלצת ציון לדו"ח מכין

אם יש צורך, לכל סטודנט/ית בנפרד

מספר הפרויקט: P-2026-078 שם הפרויקט: RF-Based Scanning and Localization System for SAR

שם המנחה : פרופ' חן אבין שם הסטודנט + ת"ז : דניאל קצ'ורה, 209156934 ; רון אשכנזי, 208282418

%	קריטריון	1 - חלש	2 - בינוני	3 - מצוין	4- ט"מ	5- מצוין
12	הגדרת המטרה - האם מטרת הפרויקט ברורה? המטרה צריכה לכלול גם את התרומה הצפויה מהשלמת הפרויקט.					
13	הבנת הבעיה - האם ניכר שהסטודנט הבין את הבעיה ואת הפתרונות הקיימים? האם נערך סקר ספרות רלוונטי? (חיפוש מקורות והבנת עבודות דומות)					
12	הארכיטקטורה המוצעת - האם הגישה הכללית ורכיבי הפתרון הוצגו והוסברו בצורה ברורה? האם ניכרת הבנה ? האם הובהר מדוע הארכיטקטורה המוצעת לפתרון ההנדסי מתאימה לפתרון ההנדסי?					
13	הצגת הדו"ח המכין – האם יש מבנה הגיוני לדו"ח המכין? האם איכות ההצגה טובה? האם ישנה התייחסות לכל חלקי הדו"ח? האם יש קשר הגיוני בין החלקים? האם רמת התחביר טובה?					
12	רמת קושי - כיצד הנך מעריך את רמת הקושי הצפויה בפרויקט בהשוואה לפרויקטים אחרים השנה או בשנים קודמות?					
13	מדדים לבדיקת הצלחת הפרויקט - האם הוגדרו קריטריונים ברורים ומדידים כמותית לבחינת הצלחת הפרויקט? האם הקריטריונים שהוצגו תואמים להגדרת מטרות הפרויקט?					
12	איכות הפרויקט המוצע - עבור פרויקטים מחקריים- מהי רמת הצעת מחקר ? עבור פרויקטים טכניים - מהי רמת המפרט טכני? האם הדרישות עונות על הצורך?					
13	המאפיינים העיקריים בפרויקט – יש לסמן את כל המאפיינים הרלוונטיים שהושגו (ולו באופן חלקי). מאפיין אחד בלבד חזכה במחצית הניקוד. שני מאפיינים ומעלה יזכו בניקוד מלא.					

הערכת רמת הקושי של הפרויקט: קל מאוד/קל/בינוני/קשה/קשה מאוד הערות: